

Classical ciphers are the best examples, and are the easiest ones as well. The problem with them is that they work on the level of ‘letters’. Classical ciphers don’t let you go below that level. In our daily life, we deal with a lot of data that’s not just a collection of letters. For example, there’s no easy way to convert an image or a video to letters. Of course, the word ‘letter’ is being used loosely here. Another question could be “Letters in which language?” Modern encoding techniques can represent letters from quite a long list of languages using clever techniques to organize bits. This leaves us with one option – abandon classical ciphers because classical ciphers cannot be used to encrypt data well enough in a digital world.

Coming to modern ciphers, we have two options: asymmetric algorithms (or public key algorithms) and symmetric algorithms (or private key algorithms). It’s well known that asymmetric algorithms provide more

security by allowing the key used for encryption to be known by anyone. However, most asymmetric algorithms are based on very complicated mathematical calculations. Though we’re going to create our own algorithm, and we would want to make a secure one, designing an asymmetric algorithm would require quite a lot of explanation and in-depth understanding of number theory, which is beyond the scope of this book.

So let’s stick to a symmetric algorithm, and move on to creating it. We’ll do this stepwise.

NOTE:

For this example that will get you started, you need to remember that this won’t be the world’s best symmetric cryptographic algorithm. The sole aim of this chapter is to show how minute details affect the overall procedure of creation and implementation of algorithms. As we move forward, we’ll show you points where you can change the variable factors of this procedure to alter the resulting algorithm. There are points that may appear to be flaws in the design process, but we’ve put these in deliberately to show you why a well thought-out method must be used, and how unnecessary complications do not necessarily make a cipher useful or strong. Nonetheless, in the end you’ll have an encryption algorithm. This algorithm will be very weak for any practical encryption (unless you only want to fool your family and friends with it).

Step 1

The first factor to be considered before starting the actual design procedure is to decide on the block and key length. The bigger both are, the better